RONIX

Project Requirements & System Design Document

SECTION 1: PROJECT OVERVIEW

RoNix is a digitized financial technology solution designed to modernize traditional group and individual micro-savings (Susu) practices. The platform transitions informal, cash-reliant savings schemes into a structured, automated, and secure digital environment. By integrating Mobile Money payment gateways (MTN, Telecel, AT), biometrics, AI assistance, and automated locked savings controls, RoNix enables individuals and groups to build financial discipline with complete transparency.

SECTION 2: CURRENT SUSU OPERATIONS & MARKET CONTEXT

Traditional Susu systems across West Africa rely on informal collectors, verbal agreements, and paper notebooks. The operational workflow currently faces severe structural friction:

- **Physical Collection:** Collectors travel to market stalls or homes daily to collect cash contributions manually.
- **Paper Ledgers:** Record-keeping is manual, leaving savings vulnerable to lost passbooks, damage, or bookkeeping errors.
- **Dispute Risks:** Payout rotation orders in traditional groups are often contested due to a lack of written rule enforcement.
- **Security & Custody Hazards:** Unregulated holding of cash exposes participants to theft, fraud, or collector default.

SECTION 3: SUSU PLANS & OFFERING ARCHITECTURE

RoNix offers two primary savings tracks tailored to individual and collective savings goals:

Plan Type	Cycle & Limits	Payout & Governance Structure
Individual Savings Plan	Daily or weekly contributions. Custom target up to 1 year (Developer UI standard capped at 30-day active cycles).	Locked savings until a designated maturity date. Auto-disbursement to linked Mobile Money or Bank account upon maturity.
Group Rotational Plan	Daily, weekly, or monthly contributions (e.g., $100 daily by 5 members). Fixed cycle length (default 30 days).	Rotational payout structure configured by Admin. Automated queue display showing current and upcoming recipients. Mandatory digital acceptance of group rules before entry.

SECTION 4: TYPES OF USERS

- **Individual Savers:** Users creating personal locked savings targets (e.g., rent, car, business capital).
- **Group Members:** Participants who join private or public rotational savings groups using generated group IDs.
- **Group Admins / Leaders:** Creators of savings groups who define rules, contribution amounts, rotation schedules, and member compliance terms.
- **System Administrators:** Platform operators monitoring system security, compliance, payment gateway settlement, and automated fee collections.

SECTION 5: CURRENT PROCESS & IDENTIFIED PROBLEMS

Identified Problem	Operational Impact
Manual Record-Keeping	Discrepancies in contribution dates, lost paper passbooks, and lack of ledger proof.
Impulse Early Withdrawals	Savers break their goals prematurely, failing to accumulate targeted funds.
Default & Rule Violations	Group members skip turns or drop out after receiving early rotational payouts.
Lack of Transaction Privacy	Public exposure of individual financial balances and personal identification.
Unreliable Payment Reconciliation	Money deducted from MoMo wallets without instant ledger crediting.

SECTION 6: PROPOSED SYSTEM SOLUTIONS

RoNix addresses existing systemic deficiencies through the following engineering implementations:

- **Strict Locked Savings Enforcer:** Disables early withdrawal access until the exact target maturity date.
- **Automated Mobile Money Integration:** Real-time collection via prompt buttons, instantly updating double-entry database ledgers.
- **Group Anonymity & Privacy Shields:** Hides full member names in public groups while restricting member rosters in private groups to authorized participants.
- **Automated Compliance & Warning System:** Tracks delayed contributions, issues automated warnings, and provides controlled removal workflows for group admins.
- **Transparent Flat-Fee Model:** Applies a transparent GHS 20 platform fee solely to individual payout recipients without distorting group total statistics.

SECTION 7: FUNCTIONAL REQUIREMENTS

1. Authentication & Identity Verification

- Sign-up via email/phone number with OTP verification, optional selfie capture, and social login support.
- Two-Factor Authentication (2FA) enforcing fingerprint or biometric/selfie verification for all financial actions.
- Password confirmation required prior to executing contribution deductions or disbursement claims.

2. Profile & Goal Setup

- Capture full name, picture, phone number, email, and preferred MoMo provider (MTN, Telecel, AT).
- Personal goal creator with target amounts, contribution frequency, and maturity dates.
- Group creation engine generating unique 10-character access codes for private groups.

3. Payment & Escrow Architecture

- Direct payment prompts for MTN, Telecel, and AT Mobile Money platforms.
- Centralized fund custody backed by a partner financial institution (Ghana Commercial Bank Escrow model) for total security and compliance.
- Webhook-driven ledger updates ensuring zero uncredited deductions or unreversed failures.

4. Locked Savings & Withdrawal Control

- Actionable "Make Withdrawal" trigger button strictly locked until maturity date.
- Auto-disbursement options transferring matured funds directly to MoMo or bank accounts.
- 24-48 hour pre-disbursement notifications and automated smart reminders.

5. AI Savings Assistant & Fraud Detection

- **Personalized Recommendations:** AI suggests contribution increments to reach targets faster.
- **Goal Prediction:** Mathematical estimation of goal completion dates based on historical payment velocity.
- **Fraud & Anomaly Monitoring:** Flags uncharacteristic activity and enforces locked ceiling limits to prevent unauthorized withdrawals.

6. Group Administration & Compliance

- Admin governance tools to set contribution amounts, rotation order, late penalties, and terms.
- Sub-admin role delegation for compliance tracking and order moderation.
- Warning issuance, non-compliance flagging, and automated countdown removal for defaulting members after cycle completion.

SECTION 8: SYSTEM USERS & CAPABILITIES

Actor	System Capabilities
Individual Member	Create personal goals, initiate MoMo contributions, track locked dates, execute 2FA, trigger matured withdrawals, view personal ledger.
Group Member	Join via code, agree to terms, track contribution calendar, view payout queue, view net payout balance after GHS 20 flat fee.
Group Admin	Configure group rules, set rotation schedule, assign sub-admins, issue compliance warnings, initiate member removal countdowns.
System Admin	Monitor escrow reconciliation, view system health reports, audit automated fee deductions (GHS 20), manage gateway API keys.

SECTION 9: SYSTEM REPORTS & ANALYTICS

- **Daily Contribution Ledgers:** Real-time reporting of all incoming MoMo deposits.
- **Payout Schedule Reports:** Calendar projection of upcoming individual and group disbursements.
- **Monetization & Fee Audit Reports:** Automated logging of GHS 20 flat service fees collected per recipient.
- **Member Compliance & Flagging Summaries:** Historical breakdown of on-time versus late contributions across groups.

SECTION 10: DATABASE ENTITIES & RELATIONSHIPS

Entity	Key Attributes & Relationships
Profile	`id`, `phone_number`, `full_name`, `momo_network`, `is_verified`. Relates 1:N to Goals & Contributions.
SavingsGoal	`id`, `user_id`, `target_amount`, `current_amount`, `target_date`, `is_locked`. Relates 1:N to Transactions.
SusuGroup	`id`, `group_code`, `admin_id`, `contribution_amount`, `frequency`, `is_private`. Relates 1:N to GroupMembers.
GroupMember	`id`, `group_id`, `user_id`, `rotation_position`, `compliance_status`. Relates 1:1 to PayoutSchedule.
Transaction	`id`, `reference_id`, `user_id`, `amount`, `type` (CONTRIBUTION/PAYOUT/FEE), `status`. Immutable audit ledger.

SECTION 11: END-TO-END SYSTEM FLOW

1. User Onboarding & 2FA / Biometric Setup

↓

2. Select Plan: Individual Locked Goal OR Group Susu

↓

3. Initiate "Make Contribution" → MoMo Prompt (MTN / Telecel / AT)

↓

4. Automated Real-Time Ledger & Escrow Deposit Credit

↓

5. Locked Savings Enforcement (Withdrawal Button Disabled Until Due Date)

↓

6. Maturity Due Date Reached → Smart Notification Sent (24-48h Prior)

↓

7. User Authenticates 2FA & Triggers "Make Withdrawal"

↓

8. Platform Deducts Flat GHS 20 Service Fee → Auto Disbursement to MoMo

SECTION 12: PROBLEM-SOLUTION MAPPING SUMMARY

Identified Problem	System Solution
Informal/Manual Order & Entry	Immutable digital PostgreSQL transaction ledger backed by Supabase.
Early Premature Withdrawals	System-enforced lock date; "Make Withdrawal" button active only on due date.
Payment Unreliability	Idempotent API webhooks verifying MoMo transactions before balance credit.
Member Non-Compliance	Compliance status flags, admin warnings, and automated cycle-end removal.
Monetization Friction	Flat GHS 20 fee billed transparently only to payout recipients at withdrawal.

SECTION 13: NON-FUNCTIONAL REQUIREMENTS

Quality	Requirement Description
Security	End-to-End Encryption, Row-Level Security (RLS) on PostgreSQL tables, mandatory 2FA biometric verification.
Reliability	Zero financial loss tolerance. Double-entry transaction logging with webhooks preventing uncredited charges.
Performance	Sub-second realtime database broadcasts for contribution updates using Supabase subscriptions.
Usability	Clean Mobile-first UX built with React Native / Expo and NativeWind, optimized for rapid MoMo interactions.
Compliance	Adherence to local Data Protection Laws and financial regulation guidelines in Ghana.

SECTION 14: OVERALL PROJECT VISION

RoNix bridges the gap between trusted community financial traditions and modern digital banking security. By combining strict locked-savings discipline, seamless Mobile Money integration, automated group rotation governance, and an intuitive cross-platform mobile experience, the platform empowers users to reach their financial goals safely, reliably, and efficiently.

SECTION 15: REPOSITORY STRUCTURE (GITHUB BLUEPRINT)

The repository architecture is structured around modular separation of concerns, housing mobile client code, database migration scripts, serverless edge functions, and project documentation deliverables (D0–D10):

```
RoNix/
├── .github/
│   ├── workflows/              # CI/CD pipelines (D4)
│   └── ISSUE_TEMPLATE/         # Task & bug tracking templates
├── docs/
│   ├── D0-Team-Charter/        # Charter, Client Intake & AI Contract
│   ├── D1-Proposal/            # Project Proposal & Feasibility Report
│   ├── D2-SRS/                 # Software Requirements Specification (RoNix PDF/MD)
│   ├── D3-Backlog-UX/          # Product Backlog, Figma links, Sprint 1 Plan
│   ├── D4-Architecture/        # Architecture Diagrams, CI/CD Baseline & Defense Deck
│   ├── D5-Sprint1-Review/      # Sprint 1 Review Report & v0.1 Notes
│   ├── D6-Sprint2-Review/      # Sprint 2 Review, Test Plan & Quality Evidence
│   ├── D7-Sprint3-Review/      # Sprint 3 Review, Security/Privacy/AI Audit & Telemetry
│   ├── D8-Sprint4-Review/      # Sprint 4 Review, Release Candidate & User Manual Draft
│   ├── D9-Final-Handoff/       # User Manual, Deployment/Ops Guide & Client Handoff Pack
│   └── D10-PostMortem/         # Repository Freeze, Post-Mortem & Peer Evaluation
├── mobile/                     # React Native / Expo Frontend Code
│   ├── app/                    # Expo Router screens & navigation
│   ├── components/             # Reusable UI elements
│   ├── services/               # API & Supabase Client configuration
│   └── package.json
├── supabase/                   # Backend Database & Edge Functions
│   ├── migrations/             # PostgreSQL DDL scripts (Profiles, Groups, Ledger)
│   ├── seed.sql                # Initial mock data for testing
│   └── functions/              # Serverless Edge Functions (Payment Webhooks, Auto-disbursement
├── tests/                      # Unit, Integration & E2E Tests (D6 Quality Evidence)
├── README.md                   # Project Root & Quickstart Guide
└── LICENSE
```

SECTION 16: BRAND IDENTITY, TYPOGRAPHY & COLOR SYSTEM

RoNix utilizes a dual-purpose design system engineered for high-contrast mobile screen legibility and high-impact physical brand merchandise (t-shirts, hoodies, and social collateral).

1. Primary Color Tokens ("Electric Wealth")

Role	Color Name	Hex Code	Application & Feel
Primary Base	Deep Midnight Navy	#0A1128	App dark background, top headers, dark t-shirt base color. Rich depth.
Primary Accent	Electric Emerald	#00E676	Main CTAs, completed contributions, positive financial balances.
Secondary Accent	Vibrant Gold / Amber	#FFC107	Payout turn highlights, compliance badges, premium alerts, merch accents.
UI Neutral Light	Soft Snow	#F8F9FA	Text on dark surfaces, app light-mode background, crisp print contrast.

Role	Color Name	Hex Code	Application & Feel
UI Neutral Dark	Slate Gray	#1E293B	Card backgrounds, subtle dividers, secondary UI text.

2. Typography & Font Stack

- **Primary UI & Data Font (Plus Jakarta Sans):** Clean geometric sans-serif optimized for mobile screens and numerical clarity in ledger tables.
- **Brand & Display Font (Space Grotesk):** Tech-forward, high-precision font for onboard headers, banners, logo typography, and apparel prints.
- **Monospace Accent (JetBrains Mono):** Reserved for transaction IDs, cryptographic references, and financial receipts.

3. Merchandise & Apparel Execution Blueprint

For company t-shirts, hoodies, and promotional collateral:

- **Base Fabric:** Heavyweight Deep Midnight Navy (#0A1128) or Jet Black.
- **Chest Branding:** RoNix minimalist icon screen-printed in Electric Emerald Ink (#00E676).
- **Back Print Slogans (Space Grotesk):**
  - *"DISCIPLINE IS THE NEW WEALTH."*
  - *"AUTOMATED SUSU. IMMUTABLE TRUST."*
  - *"SAVINGS ON LOCK. PAYOUT ON TIME."*